Cover letter

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Unsent submission correspondence; prepared for author review.

Dear Editors,

Please consider the manuscript “Residual-dimension obstructions for null-projected supersymmetric Yang-Mills theories” by Deep Bhattacharjee.

The paper proves necessary residual-dimension and signature conditions for null-projected supersymmetric Yang-Mills constructions under five explicit axioms. It also gives a separate unitary helicity-span obstruction. It does not assert existence in every allowed signature or a nonperturbative quantum construction.

A detailed first-principles obstruction theorem for supersymmetric gauge constructions fits the journal’s fundamental theoretical-physics scope and long-form treatment.

The prior public preprint is `https://hexagonmath.org/2610.00471v1`. The research manuscript/source archive is `https://doi.org/10.5281/zenodo.23611588`. Editable sources and verification records are supplied with the manuscript.

No general Lean formalisation is claimed. The proof is written in the manuscript; the gamma-matrix calculations are supplementary finite numerical checks.

The manuscript and the accompanying declaration disclose the substantive use of Claude and OpenAI tools. The submitting author has declared that all parts have been checked and understood by a human. The named human authors retain responsibility for the content.

Sincerely,

Deep Bhattacharjee
Corresponding author
itsdeep@live.com
ORCID: 0000-0003-0466-750X